IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2023

DIGITAL CONTROL SYSTEMS

Tuesday 9 May 10:00 am

Time allowed: 3:00 hours

There are THREE questions on this paper.

Answer ALL questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : D. Angeli

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DIGITAL CONTROL SYSTEMS

Information for candidates:

- For a given signal r (also indicated as $r(k)$, $r(kT)$ or $r(t)$), $R(z)$ denotes its Z-transform. All signals are assumed to be zero for negative times.
- $Z\left(\frac{1}{s}\right) = Z(1(t)) = Z(1(kT)) = \frac{z}{z-1}$
- $Z\left(\frac{1}{s+a}\right) = Z\left(e^{-at}\right) = Z\left(e^{-akT}\right) = \frac{z}{z-e^{-aT}}$
- $Z\left(\frac{1}{s^2}\right) = Z(t) = Z(kT) = \frac{zT}{(z-1)^2}$
- $Z\left(a^k\right) = \frac{z}{z-a}$
- $Z\left(ka^{k-1}\right) = \frac{z}{(z-a)^2}$
- Transfer function of the ZOH: $H_0(s) = \frac{1-e^{-sT}}{s}$
- Tustin transformation: $s = \frac{2}{T}\frac{z-1}{z+1}$

1. Consider the digital control system in Figure 1.1.

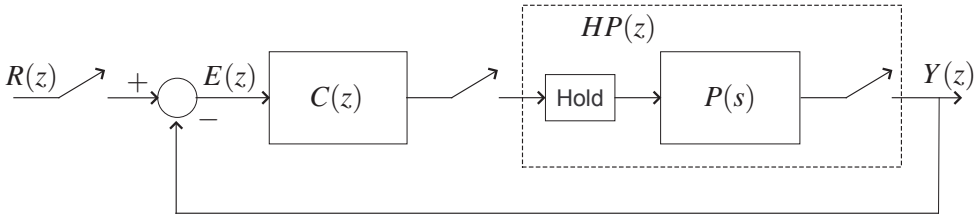


Figure 1.1 Block diagram for question 1.

Let

$$P(s) = \frac{1}{4s+1}.$$

Assume the hold is a zero-older hold (ZOH) and let the sampling period be $T = 1$ sec.

a) Compute the equivalent discrete-time model $HP(z)$ for the plant interconnected to the hold and the sampler. [4 marks]

b) Consider a continuous-time controller described by the transfer function

$$C(s) = \frac{as+b}{s}$$

with $a > 0$ and $b > 0$ parameters to be selected. Discretize the controller $C(s)$ using the Tustin transformation. Compute explicitly the resulting discrete-time controller. Determine for which values of a and b the controller has a zero inside the unity disk (in this case we say that the controller is minimum phase). [4 marks]

c) Using the results of parts a) and b) compute the closed-loop transfer function from the input $R(z)$ to the output $Y(z)$. [4 marks]

d) The resulting characteristic polynomial is

$$z^2 + Az + B$$

with

$$A = \tfrac{1}{2}\left[(1-e^{-\frac{1}{4}})(2a+b) - 2(1+e^{-\frac{1}{4}})\right],$$
$$B = \tfrac{1}{2}\left[2e^{-\frac{1}{4}} + (1-e^{-\frac{1}{4}})(b-2a)\right].$$

Study the stability properties of the discrete-time closed-loop system finding what conditions a and b must satisfy to have an asymptotically stable closed-loop system and a minimum phase controller. [8 marks]

2. Consider the digital control system in Figure 2.1.

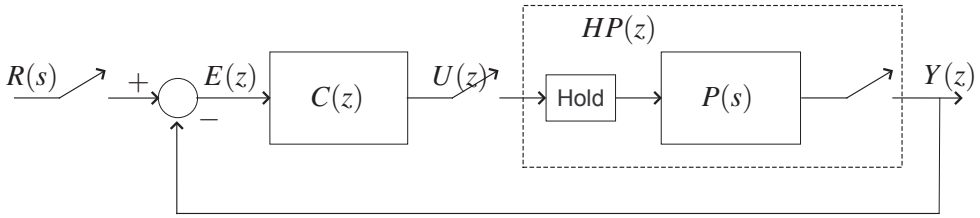


Figure 2.1 Block diagram for question 2.

The continuous-time system is defined by

$$\begin{aligned} \dot{x}(t) &= Fx(t)+Gu(t), \\ y(t) &= Lx(t), \end{aligned} \tag{2.1}$$

with

$$F=\begin{bmatrix} 0 & 1 \\ 0 & -1 \end{bmatrix}, \qquad G=\begin{bmatrix} 0 \\ 1 \end{bmatrix}, \qquad L=\begin{bmatrix} 1 & 0 \end{bmatrix}.$$

Let the sampling time be $T=1$ sec.

a) Discretize the continuous-time state-space equations (2.1). To this end, determine the matrices A, B and C of the representation

$$\begin{aligned} x(k+1) &= Ax(k)+Bu(k), \\ y(k) &= Cx(k), \end{aligned} \tag{2.2}$$

when the input $u(t)$ is sampled and fed to a zero-order hold.
Hint: recall that $e^{Ft}=\mathscr{L}^{-1}\left[(sI-F)^{-1}\right]$, *where* $\mathscr{L}^{-1}$ *indicates the inverse Laplace transform operator.*

[4 marks]

b) Compute the input-output transfer function of the discrete-time system (2.2) from the matrices A, B and C.

[3 marks]

c) The continuous-time state-space system (2.1) has input-output transfer function

$$\frac{Y(s)}{U(s)}=P(s)=L(sI-F)^{-1}G=\frac{1}{s(s+1)}.$$

Compute the equivalent discrete-time model $HP(z)$ for the plant $P(s)$ interconnected to the zero-order hold and the sampler. Show that $HP(z)$ is the same transfer function as the one obtained in part b).

[3 marks]

d) Consider system (2.2) obtained in part a). Determine a state-feedback control $u(k)=-Kx(k)$ such that the closed-loop response is deadbeat (all closed-loop eigenvalues are zero). Note that Ackermann's formula is not your quickest option.

[4 marks]

e) The resulting closed-loop system obtained with the control found in part e) is

$$x(k+1)=(A-BK)x(k)=\begin{bmatrix} 0.418 & 0.175 \\ -1 & -0.418 \end{bmatrix}x(k).$$

Show that for any initial condition $(x_1(0), x_2(0))$, we have that $x_1(2)=0$ and $x_2(2)=0$ (for your computations in this question keep the approximation to the third decimal digit). [3 marks]

f) Consider the equivalent discrete-time model $HP(z)$ computed in part d). Determine a controller described by an equation of the form

$$C(z) = \frac{s_0 z^m + s_1 z^{m-1} + \cdots + s_{m-1} z + s_m}{z^m + v_1 z^{m-1} + \cdots + v_{m-1} z + v_m}$$

such that the closed-loop response is deadbeat (all poles are at zero). Select m such that $C(z)$ has the smallest possible order. There is no need to give the numerical values of the coefficient s_i and v_i, just write the equations that relate these quantities with one another (preferably in matrix form).

[3 marks]

3. a) Consider the continuous-time plant

$$P(s) = \frac{\frac{1}{2}}{(s^2+\frac{1}{4})(s^2+\frac{9}{4})},$$

connected to a ZOH and a sampler.

i) Determine the poles of $P(s)$ and discuss the stability of the continuous-time system described by $P(s)$. [3 marks]

ii) For the generic sampling time T, determine $HP(z)$. Since determining $HP(z)$ could become computationally demanding, it is advised that you stop your computation once you find a form of the type

$$HP(z) = \alpha_0 + (z-1)\left(\frac{\alpha_1 z+\alpha_2}{z^2+\alpha_3 z+\alpha_4} + \frac{\alpha_5 z+\alpha_6}{z^2+\alpha_7 z+\alpha_8}\right),$$

where you need to determine the real-valued coefficients α_i, $i = 0,\ldots,8$. [4 marks]

iii) Set $T = \pi$ sec and compute $HP(z)$ for this specific choice of T. Moreover, comment on whether the choice $T = \pi$ sec is implementable in practice. [3 marks]

iv) Compute the poles of $HP(z)$ determined in part iii) and explain why the number of poles is lower than the number of poles of the continuous-time transfer function $P(s)$. [4 marks]

b) Consider now the function $X(z)$ given by

$$X(z) = \frac{z^2-2z}{z^2-2z+1}x(0) + \frac{1}{z-1}U(z),$$

with input

$$U(z) = \frac{z}{(z-\frac{1}{2})^2}.$$

i) Compute $x(k)$. [3 marks]

ii) Assess the convergence properties of $x(k)$ as $k \to \infty$ as a function of $x(0)$. [3 marks]